Jtop tool

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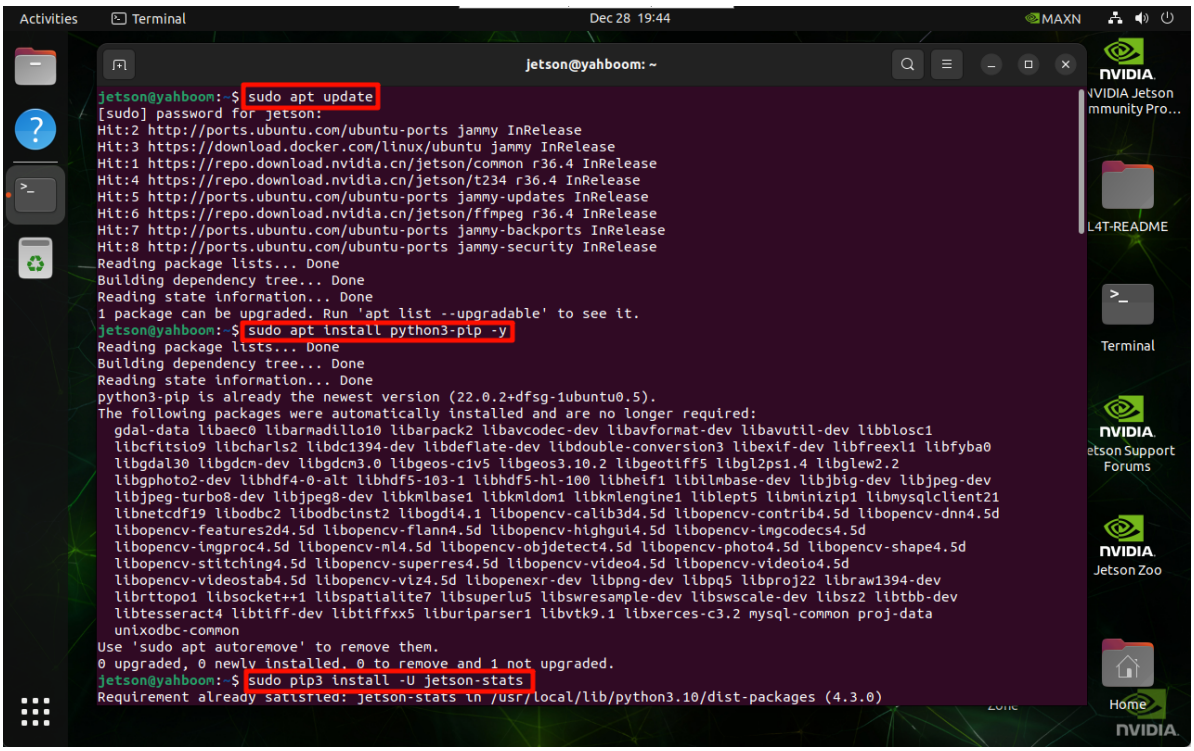
Jtop is a system monitoring tool developed for NVIDIA Jetson series devices. It can display the resource usage of various aspects of Jetson devices, such as CPU, GPU, memory, disk, network, etc., and can display different hardware temperatures, power consumption, frequency, etc. in real time.

1. Install Jtop

```
sudo apt update
sudo apt install python3-pip -y
sudo pip3 install -U jetson-stats
```



2. Best performance mode

2.2. Enable MAX mode

Enabling MAX Power Mode on Jetson will ensure that all CPU and GPU cores are turned on:

```
sudo nvpmodel -m 0
```

2.2. Enable Jetson Clocks

Enabling Jetson Clocks will ensure that all CPU and GPU cores are running at maximum frequency:

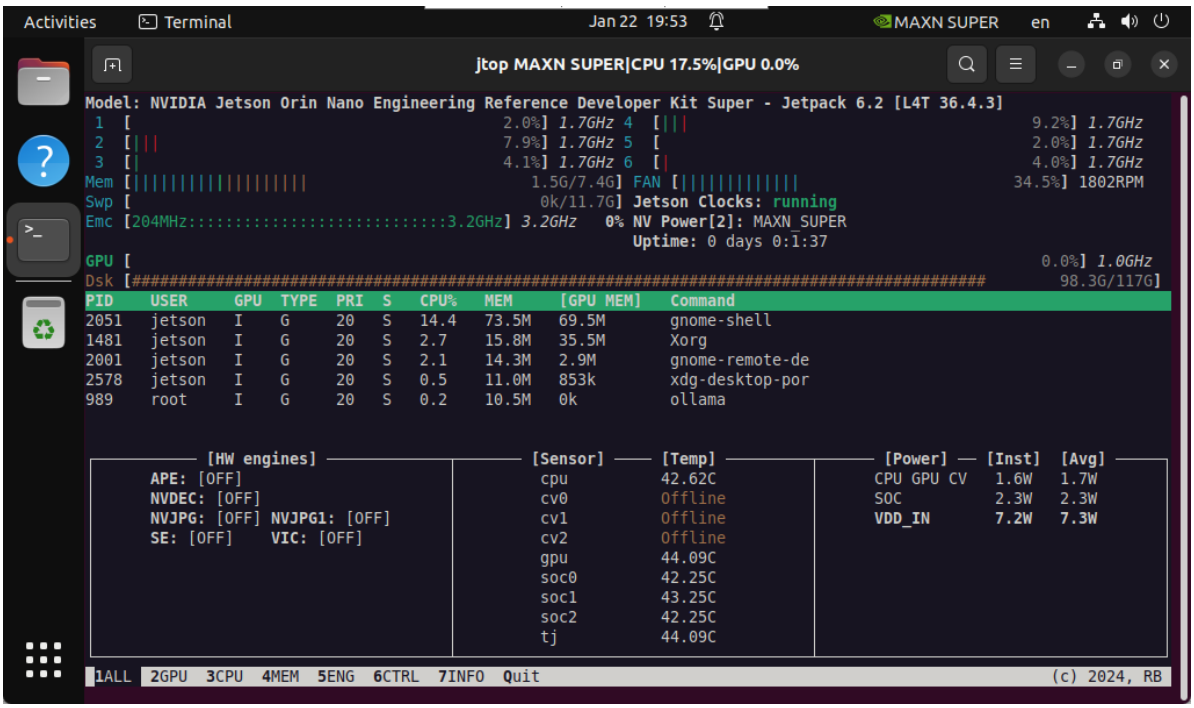
```
sudo jetson_clocks
```

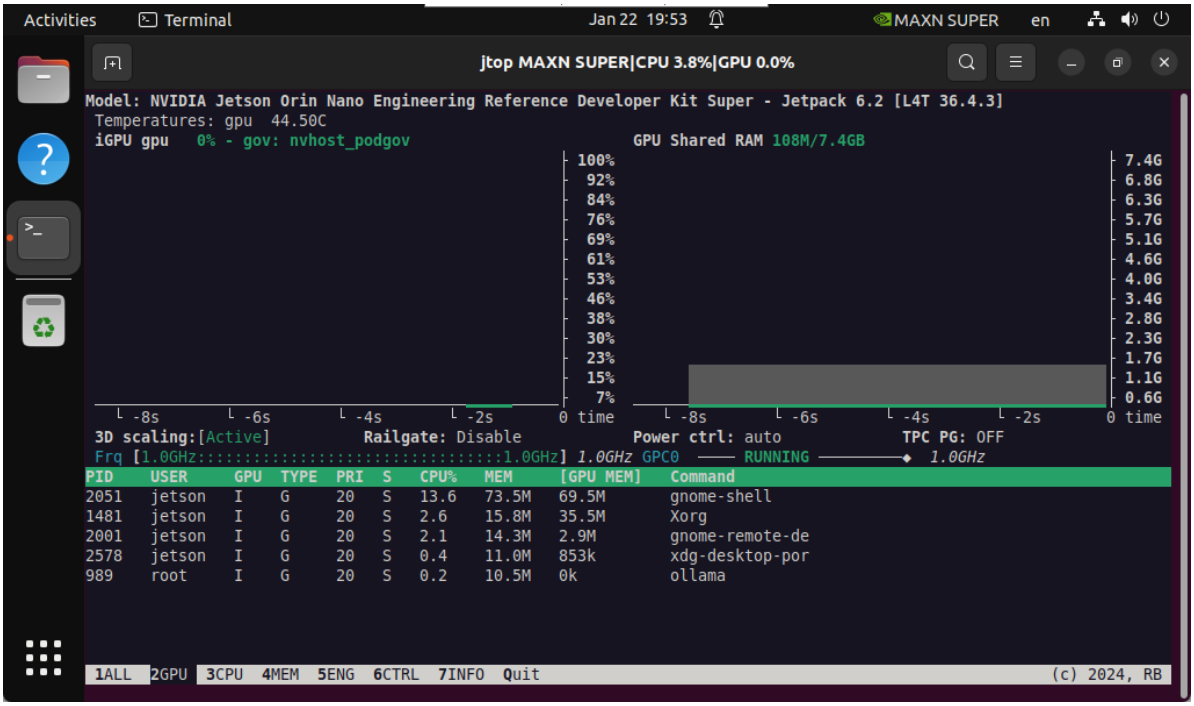
3. Use Jtop

Only after restarting the system can you enter the jtop command in the terminal to start the Jtop tool:

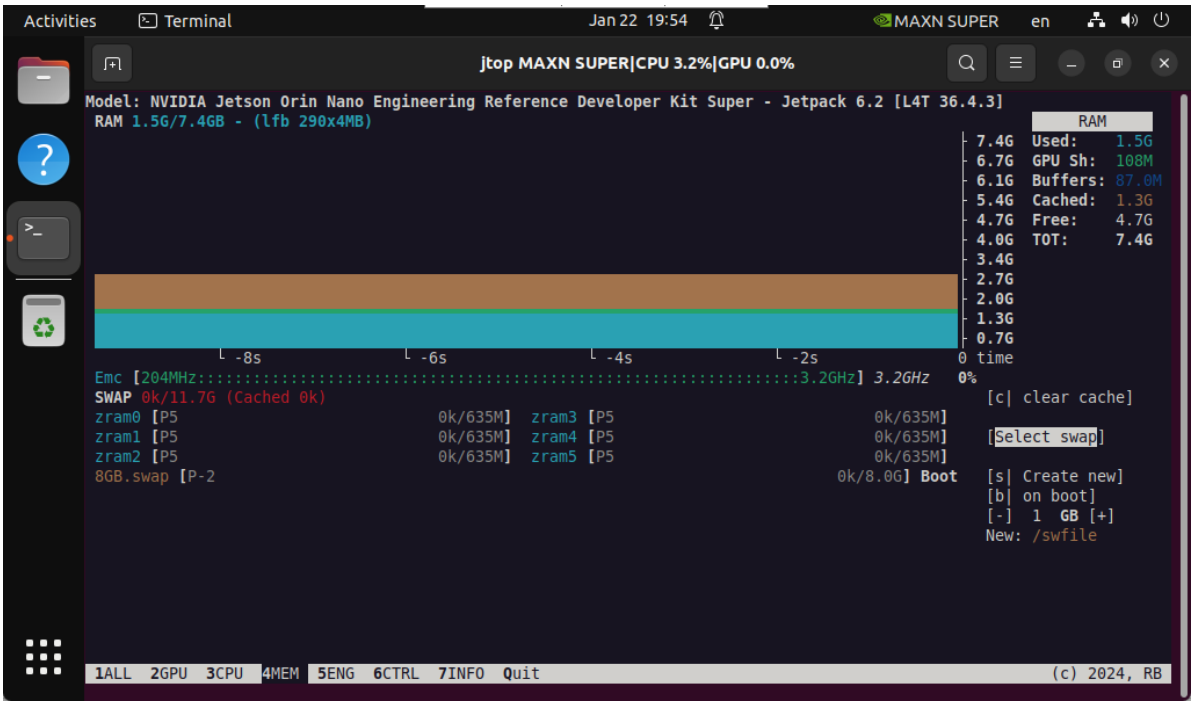
```
jtop
```

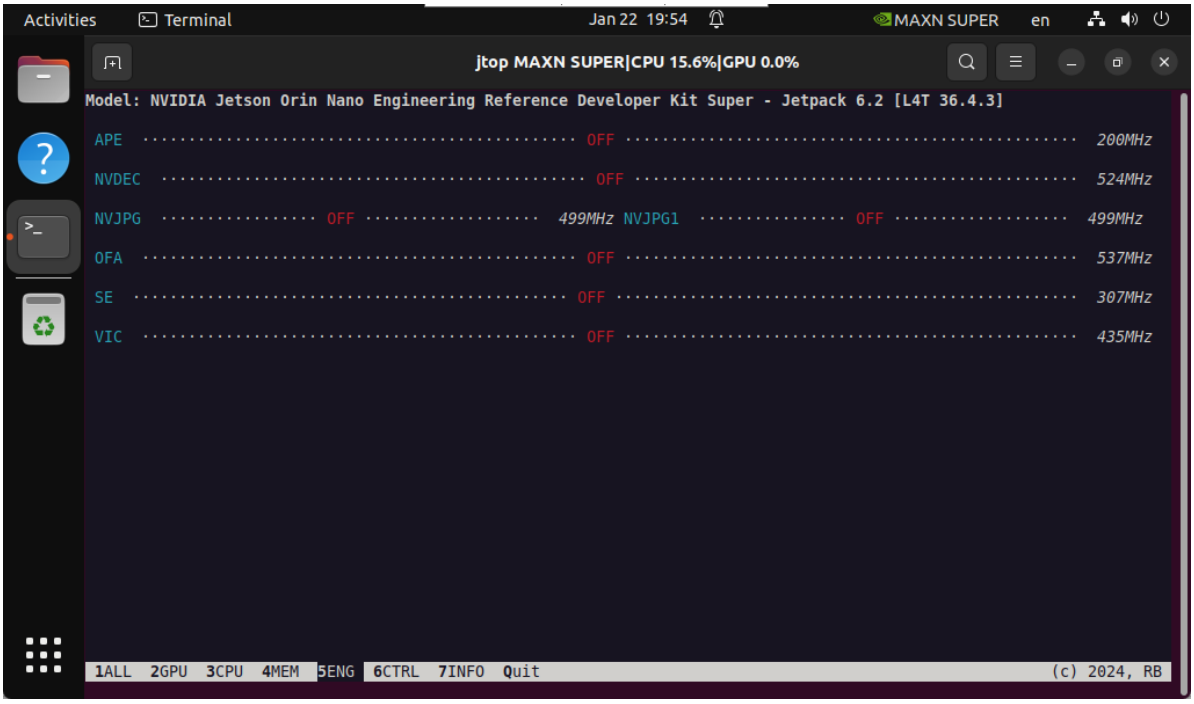
Note: Only when the motherboard power mode is selected as MAXN will the strongest performance parameters be displayed!

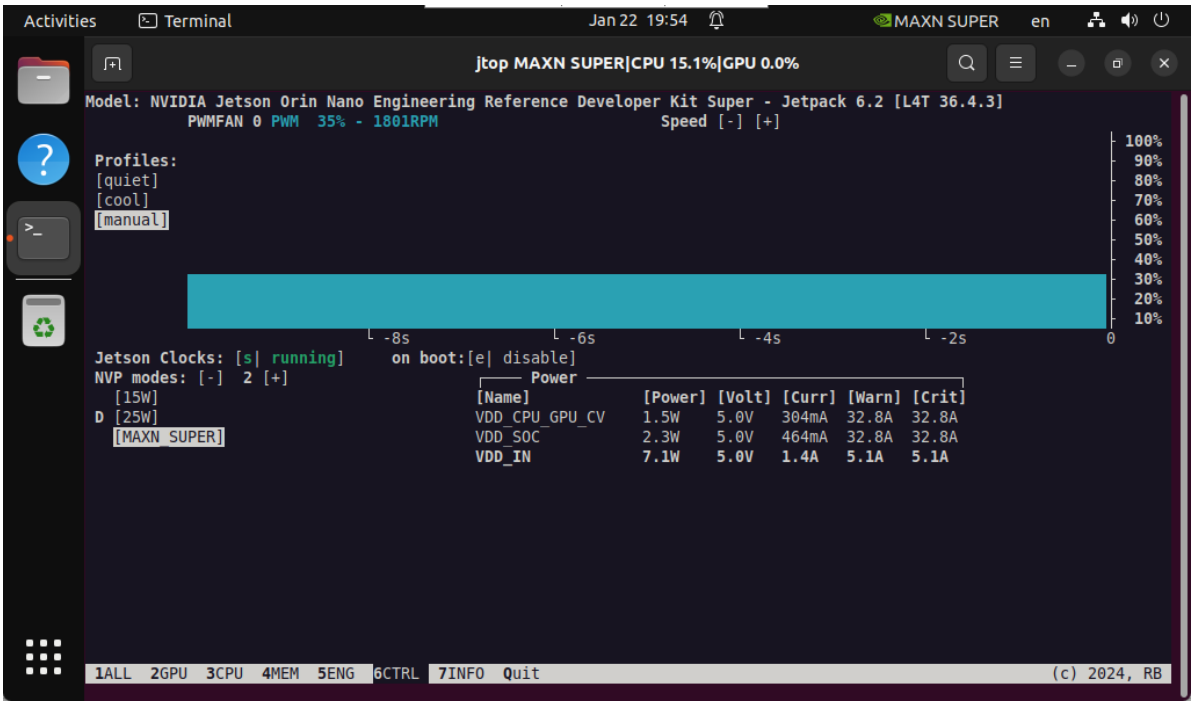


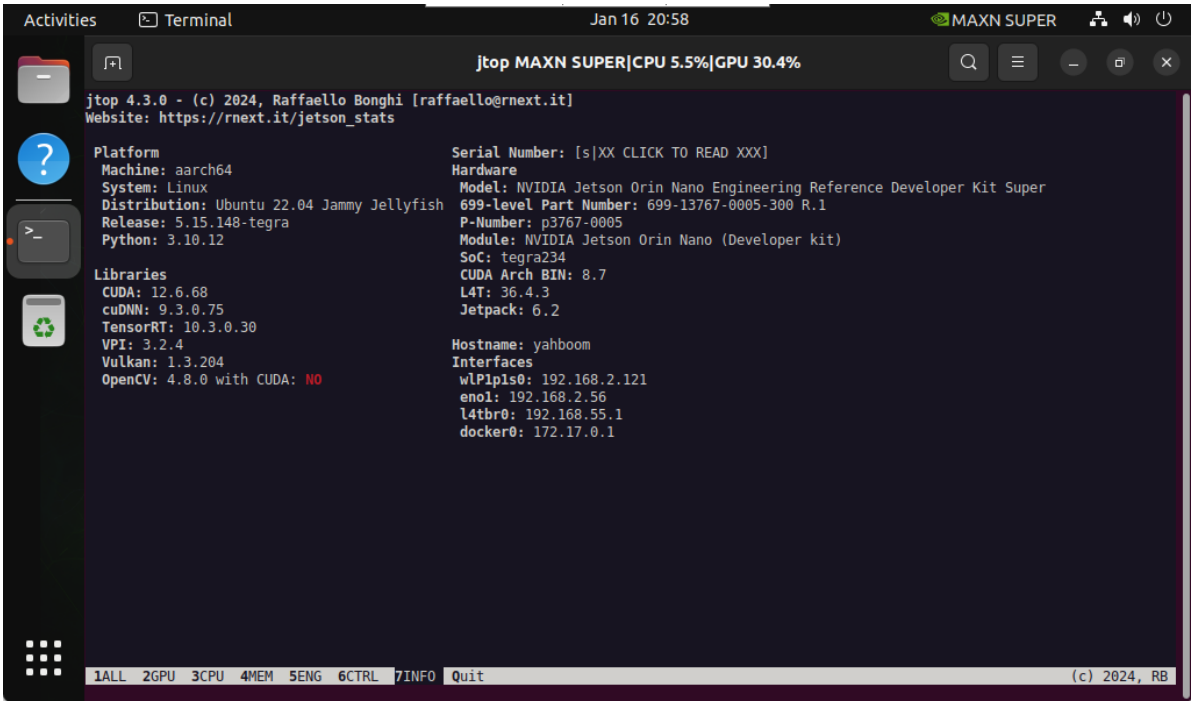


Activities
Terminal
Jan 22 19:53
MAXN SUPER
en
jtop MAXN SUPER|CPU 16.4%|GPU 0.0%
Model: NVIDIA Jetson Orin Nano Engineering Reference Developer Kit Super - Jetpack 6.2 [L4T 36.4.3]
ALL [|||||||||||||| 15.3%]
1 1% Schedutil
4 7% Schedutil
2 6% Schedutil
5 3% Schedutil
3 2% Schedutil
6 0% Schedutil
ARMv8 Processor rev 1 (v8l)
Frq [1.7GHz::::::::::::::::::::::::::::::1.7GHz] 1.7GHz
1ALL 2GPU 3CPU 4MEM 5ENG 6CTRL 7INFO Quit
(c) 2024, RB


Activities
Terminal
Jan 22 19:54
MAXN SUPER
en
jtop MAXN SUPER|CPU 3.2%|GPU 0.0%
Model: NVIDIA Jetson Orin Nano Engineering Reference Developer Kit Super - Jetpack 6.2 [L4T 36.4.3]
RAM 1.5G/7.4GB - (lfb 290x4MB)
RAM
Used: 1.5G
GPU Sh: 108M
Buffers: 87.0M
Cached: 1.3G
Free: 4.7G
TOT: 7.4G
Emc [204MHz::3.2GHz] 3.2GHz 0%
SWAP 0k/11.7G (Cached 0k)
zram0 [P5 0k/635M]
zram1 [P5 0k/635M]
zram2 [P5 0k/635M]
zram3 [P5 0k/635M]
zram4 [P5 0k/635M]
zram5 [P5 0k/635M]
8GB.swap [P-2 0k/8.0G] Boot
[c| clear cache]
[Select swap]
[s| Create new]
[b| on boot]
[-] 1 GB [+]
New: /swfile
1ALL 2GPU 3CPU 4MEM 5ENG 6CTRL 7INFO Quit
(c) 2024, RB


Activities
Terminal
Jan 22 19:54
MAXN SUPER
en
jtop MAXN SUPER|CPU 15.6%|GPU 0.0%
Model: NVIDIA Jetson Orin Nano Engineering Reference Developer Kit Super - Jetpack 6.2 [L4T 36.4.3]
APE OFF 200MHz
NVDEC OFF 524MHz
NVJPG OFF 499MHz NVJPG1 OFF 499MHz
OFA OFF 537MHz
SE OFF 307MHz
VIC OFF 435MHz
1ALL 2GPU 3CPU 4MEM 5ENG 6CTRL 7INFO Quit
(c) 2024, RB


Activities
Terminal
Jan 22 19:54
MAXN SUPER
en
jtop MAXN SUPER|CPU 15.1%|GPU 0.0%
Model: NVIDIA Jetson Orin Nano Engineering Reference Developer Kit Super - Jetpack 6.2 [L4T 36.4.3]
PWMFAN 0 PWM 35% - 1801RPM
Speed [-] [+]
Profiles:
[quiet]
[cool]
[manual]
100%
90%
80%
70%
60%
50%
40%
30%
20%
10%
-8s
-6s
-4s
-2s
0
Jetson Clocks: [s| running] on boot:[e| disable]
NVP modes: [-] 2 [+]
[15W]
D [25W]
[MAXN_SUPER]
Power
[Name] [Power] [Volt] [Curr] [Warn] [Crit]
VDD_CPU_GPU_CV 1.5W 5.0V 304mA 32.8A 32.8A
VDD_SOC 2.3W 5.0V 464mA 32.8A 32.8A
VDD_IN 7.1W 5.0V 1.4A 5.1A 5.1A
1ALL 2GPU 3CPU 4MEM 5ENG 6CTRL 7INFO Quit
(c) 2024, RB


Activities
Terminal
Jan 16 20:58
MAXN SUPER
jtop MAXN SUPER|CPU 5.5%|GPU 30.4%
jtop 4.3.0 - (c) 2024, Raffaello Bonghi [raffaello@rnext.it]
Website: https://rnext.it/jetson_stats
Platform
Machine: aarch64
System: Linux
Distribution: Ubuntu 22.04 Jammy Jellyfish
Release: 5.15.148-tegra
Python: 3.10.12
Libraries
CUDA: 12.6.68
cuDNN: 9.3.0.75
TensorRT: 10.3.0.30
VPI: 3.2.4
Vulkan: 1.3.204
OpenCV: 4.8.0 with CUDA: NO
Serial Number: [s|XX CLICK TO READ XXX]
Hardware
Model: NVIDIA Jetson Orin Nano Engineering Reference Developer Kit Super
699-level Part Number: 699-13767-0005-300 R.1
P-Number: p3767-0005
Module: NVIDIA Jetson Orin Nano (Developer kit)
SoC: tegra234
CUDA Arch BIN: 8.7
L4T: 36.4.3
Jetpack: 6.2
Hostname: yahboom
Interfaces
wlP1p1s0: 192.168.2.121
eno1: 192.168.2.56
l4tbr0: 192.168.55.1
docker0: 172.17.0.1
1ALL 2GPU 3CPU 4MEM 5ENG 6CTRL 7INFO Quit
(c) 2024, RB